MINOR IN MATHEMATICS

Rationale

The CMS Department offers a Mathematics minor for RHU students. It is designed to give students a solid foundation in mathematics and some advanced-level experience in the discipline.

Program Objectives

The aims of a minor in Mathematics are:

- Provide RHU graduates with a basic proficiency in Mathematics to compete at the university level.
- Support RHU graduates with essential mathematical skills to enhance their knowledge and understanding of their majors.

Learning Outcomes

At the end of this minor, the student is expected to demonstrate:

- An ability to analyze mathematically experimental or physical results
- An ability to use mathematical techniques, skills, and facts in their respective research

Entrance Requirements

Interested RHU students must complete the appropriate form declaring their intent to minor in Mathematics while completing their regular major.

Mathematics Minor Policies

- Obtain a Minor Cumulative Grade Point Average of no less than 70 %.
- Up to four courses between student major requirements and mathematics minor requirements are counted to fulfill the mathematics minor requirements

Career Options

This minor allows its holders to pursue careers across various sectors, regardless of their major. Graduates of this minor can seek jobs in teaching, banking and finance, computing, and statistics.

Program Requirements

To complete the Minor in Mathematics, a student must

1) Declare a Minor in Mathematics by completing the Minor Declaration Form;
2) Obtain the approval of the dean of the college major and the dean of CAS
3) Obtain a Minor Cumulative Grade Point Average of no less than 70%.
4) Complete 18 credits of Mathematics coursework as specified below.

It should be noted that the student must complete at least six credit hours of coursework that are not counted toward the requirements for their major or any other minor.

Curriculum and Program

The mathematics minor consists of six courses (18 credits), of which three are mandatory and three are electives, selected to satisfy the requirements of the proposed program objectives and learning outcomes.

I. Mandatory Courses

The three mandatory courses (9 Cr.) are:

Course #	Title	Credits	Prerequisites
MATH 215	Calculus III	3	
MATH 216	Calculus IV	3	MATH 215
MATH 311	Linear Algebra with Applications	3	

II. Elective Courses

The Mathematics Minor Advisor will choose the elective courses based on students' needs and backgrounds. Students may choose three elective courses (9 credits) from the following list.

Course #	Title	Credits	Prerequisites
MATH 210	Discrete Mathematics	3	
MATH 217	Advanced Calculus	3	MATH 215
MATH 314	Ordinary Differential Equations	3	MATH 216
MATH 316	Introduction to Analysis	3	MATH 217
MATH 317	Partial Differential Equations	3	MATH 314
MATH 318	Vector Calculus	3	MATH 217 & MATH 311
MATH 351	Probability and Statistics	3	MATH 215
MATH 421	Numerical Analysis	3	MATH 311
MATH 425	Introduction to Complex Variables	3	MATH 217
MATH 442	Introduction to Graph Theory	3	MATH 316, MATH 210

Courses Description

MATH 210	Discrete Mathematics	3(3,0)
Logic, propositional equivalences, predicates and quantifiers, methods of proof, proof strategy, mathematical induction, recursive definitions, strong induction, structural induction, sets and set operations, functions, growth of functions, relations and their properties, representing relations, equivalence relations, introduction to graphs, and graph terminology.		

MATH 215	Calculus III	3(3,0)
This course covers complex numbers, trigonometric functions, inverse trigonometric functions, hyperbolic functions, and integration techniques. The course also covers infinite sequences and series: limits of sequences of numbers, bounded sequences, integral test for series, comparison tests, ratio and root tests, alternating series, absolute and conditional convergence, power series, Taylor and MacLaurin series, and applications of power series. Polar functions, polar coordinates, and graphing of polar curves are also covered. In addition, topics from multivariable calculus are introduced: double integrals, applications to double integrals, and double integrals in polar form.		

MATH 216	Calculus IV	3(3,0)
Multivariable calculus and vector analysis for three-dimensional problems. Topics include 3-D vectors, dot and cross products; quadratic surfaces; limits, continuity, and partial derivatives; gradient, chain rule, and directional derivatives; tangent planes, differentials, extrema, and saddle points; triple integrals in rectangular, cylindrical, and spherical coordinates with applications to mass and moments; vector fields, work, flux, and Green's Theorem; and an introduction to Fourier series.		

MATH 311	Linear Algebra with Applications	3(3,0)
Systems of linear equations, matrix algebra, linear transformations, determinants, vector spaces, eigenvalues and eigenvectors, symmetric matrices, orthogonality, diagonalization.		

MATH 314	Ordinary Differential Equations	3(3,0)
First-order linear differential equations; linear differential equations of second and higher order; linear differential equations with variable coefficients; series solutions; systems of differential equations; Laplace transforms and their inverses. **Prerequisite:** MATH 216.		

MATH 317	Partial Differential Equations	3(3,0)
Lagrange theorem, boundary conditions of first-order equations, non-linear first-order PDEs, Charpit's equations, second-order PDEs, classification: hyperbolic, parabolic, and elliptic, the method of separation of variables, introduction to Fourier series and integrals, boundary value problems: heat equation, wave equation, Laplace equation, and finite-length strings. **Prerequisite**: MATH 314.		

MATH 351	Probability and Statistics	3(3,0)
Probability and conditional probability; discrete and continuous random variables; marginal distributions; expectation, variance, mean, median, covariance, and correlation; conditional expectation; Normal distribution; sampling distribution; prediction and confidence intervals; hypothesis testing; and regression line and correlation coefficients. **Prerequisite**: MATH 215		

MATH 421	Numerical Analysis	3(3,0)
Error Analysis, solutions of nonlinear equations using fixed-point Newton-Raphson-Muller's methods, solution of linear systems using Gaussian elimination-iterative methods, interpolation and approximation using Taylor series-Lagrange approximation-Newton polynomials, numerical differentiation and integration, numerical optimization, solutions of ordinary and partial differential equations using Euler's and Heun's and Rung-Kutta methods, MATLAB codes related to the topics mentioned above. **Prerequisite**: MATH 311.		

MATH 316	Introduction to Analysis	3(3,0)
Ordered, finite, countable and uncountable sets, sequences, subsequences, Cauchy sequences, upper and lower limits, series, limits of sequences of functions, continuity and compactness, connectedness, infinite limits, and limits at infinity, differentiation of vector-valued functions, series of functions, uniform convergence and continuity, functions of several variables, the inverse function and the implicit function theorems, the rank theorem. **Prerequisite**: MATH 217.		

MATH 318	Vector Calculus	3(3,0)
Theory of vector-valued functions, divergence, gradient, curl, vector fields, path integrals, surface integrals, constrained extrema, and Lagrange multipliers. Implicit function theorem. Green's and Stokes' theorems: an introduction to differential geometry. **Prerequisites**: MATH 217 and MATH 311.		

MATH 425	Introduction to Complex Variables	3(3,0)
Complex numbers, geometric representation, analytic functions, real line integrals, complex integration, power series, residues, poles, and conformal mappings. **Prerequisite**: MATH215.		

MATH 442	Introduction to Graph Theory	3(3,0)
Combinatorics through graph theory. Topics include connectedness, factorization, Hamiltonian graphs, network flows, Ramsey numbers, graph coloring, graph automorphisms, and Pólya's Enumeration Theorem—prerequisites: MATH 316 and Math 210.		